

Department of Mathematical and Computational Sciences
National Institute of Technology Karnataka, Surathkal

Assignment 1 – MA725

Submission Date: _____

Instructions:

- Answer all questions.
 - Show all necessary calculations clearly.
 - Write neatly and clearly.
 - **Do not copy from each other. Plagiarism will result in zero marks.**
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1. Solve the 3×3 linear system

$$x_1 + 2x_2 + 3x_3 = 1,$$

$$2x_1 + x_2 + 4x_3 = 1,$$

$$3x_1 + 4x_2 + x_3 = 1,$$

using a simple (easily programmable) algorithm.

2. Consider the vector $\mathbf{x} = [1, -2, 3]^T$. Compute $\|\mathbf{x}\|_p$ for $p = 1, 2, \infty$.

3. Let

$$A = \begin{bmatrix} 4 & 2 & 2 \\ 2 & 4 & 2 \\ 2 & 2 & 4 \end{bmatrix}, \quad \text{show that } A \text{ is positive definite.}$$

4. If $A \in \mathbb{R}^{n \times n}$ is a positive definite matrix, then

(a) A is non-singular.

(b) $a_{ii} > 0$ for $i = 1, 2, \dots, n$.

(c) $\max_{1 \leq k \leq n} |a_{kj}| \leq \max_{1 \leq i \leq n} |a_{ii}|$.

(d) $a_{ij}^2 < a_{ii}a_{jj}$ for $i \neq j$.

5. Vector norms must also satisfy $\|cx\| = |c| \|x\|$. The norm must be positive except when $x = 0$. Which of these are norms for vectors $x = (x_1, x_2) \in \mathbb{R}^2$?

$$\|x\|_A = |x_1| + 2|x_2|, \quad \|x\|_B = \min\{|x_1|, |x_2|\},$$

$$\|x\|_C = \|x\|_1 + \|x\|_\infty, \quad \|x\|_D = \|Ax\| \quad (\text{this answer depends on } A).$$

6. Use $\|AA^{-1}\| \leq \|A\| \|A^{-1}\|$ to prove that the condition number is at least 1.

7. Determine the bandwidth of the following matrix:

$$A = \begin{bmatrix} 1 & 2 & 0 & 0 & 0 \\ 3 & 4 & 5 & 0 & 0 \\ 0 & 6 & 7 & 8 & 0 \\ 0 & 0 & 9 & 0 & 1 \\ 0 & 0 & 0 & 2 & 3 \end{bmatrix}$$